

FRANCESCO BARILE

CONTACT INFORMATION

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ACADEMIC POSITIONS

University of Milano-Bicocca, Milan, Italy

• **Postdoctoral Research Fellow**

Mar 2025 - present

(*Assegno di Ricerca*) pursuant to Article 22 of Italian Law No. 240/2010

Department of Economics, Management and Statistics

Project: Joint Bayesian Learning of Related Clinical Studies [\[link\]](#)

Project PI: Prof. Bernardo Nipoti, University of Milano-Bicocca

Partner unit coordinator: Prof. Francesco C. Stingo, University of Florence

Funding: Fondazione Cariplo

EDUCATION

University of Milano-Bicocca, Milan, Italy

• **Ph.D. in Statistics**

2020 - 2024

Department of Economics, Management and Statistics

Defense date: Feb 14th, 2025

Thesis title: “Advances in Bayesian Methods:

Final evaluation: Excellent

Nonparametric Network Modeling and Scalable Computations”

Supervisor: Prof. Bernardo Nipoti

University of Calabria, Rende, Italy

• **M.Sc. in Finance and Insurance** (English)

2016 - 2019

Department of Economics, Statistics and Finance

Thesis title: “Estimating Stochastic Models for commodity futures”

110/110 cum laude

Supervisor: Prof. Arturo Leccadito

• **B.Sc. in Statistics for Business and Insurance**

2012 - 2015

Department of Economics, Statistics and Finance

Thesis title: “Test di Wilcoxon: applicazioni in ambito medico-farmaceutico”

110/110 cum laude

Supervisor: Prof. Agostino Tarsitano

RESEARCH VISITS

• **Visiting Researcher**

Nov 2023 - Aug 2024

Lancaster University, Lancaster, United Kingdom

School of Mathematical Sciences

Supervisor: Prof. Christopher Nemeth

• **Visiting Graduate Student (Research Only)**

Aug - Dec 2018

University of Western Ontario, London, Canada

RESEARCH INTERESTS

Bayesian nonparametrics, Network modeling, Computational statistics, Bayesian statistics

SCHOLARSHIPS AND AWARDS

•Junior Travel Support Award	2026
ISBA 2026, Nagoya, Japan	
•Junior Travel Support Award	2025
BayesComp 2025, Singapore	
•Exchange Extra-UE Funded Mobility Programme Award (10 months)	2023
University of Milano-Bicocca	
•Ph.D. Scholarship Award (4 years)	2020
University of Milano-Bicocca	
•Overseas Research Scholarship Award (4 months)	2018
University of Calabria	

PUBLICATIONS

REFERRED JOURNAL ARTICLES

[1] **Barile, F.**, Lunagómez, S., & Nipoti, B. (2025). Bayesian nonparametric modeling of heterogeneous populations of networks. *Bayesian Analysis*. Advance Publication. DOI: [10.1214/26-BA1588](https://doi.org/10.1214/26-BA1588). Code: [\[link\]](#)

REFERRED CONFERENCE PROCEEDINGS

[1] **Barile, F.**, Lunagómez, S., & Nipoti, B. (2023) Flexible modelling of heterogeneous populations of networks: a Bayesian nonparametric approach. In C. Rampichini, M. La Rocca, P. Coretto, G. Giordano, M.L. Parrella (Eds.). *Book of Abstracts and Short Papers CLADAG 2023* (p. 69). Pearson. ISBN: 978-88-9193-563-2. [\[link\]](#)

[2] **Barile, F.**, Lunagómez, S., & Nipoti, B. (2025) Clustering multiple network data with an application to the EU food trade market. In: A. Pollice, P. Mariani (Eds.). *Methodological and Applied Statistics and Demography II: SIS 2024, Short Papers, Solicited Sessions* (pp. 68–72). Springer. ISBN: 978-3-031-64350-7. DOI: [10.1007/978-3-031-64350-7](https://doi.org/10.1007/978-3-031-64350-7).

[3] **Barile, F.**, Dallari, S., Pacifici, C., & Grün, B. (2025). Assessing neuron response to external stimuli with a data-driven procedure for spike train extraction and GAMLSS regressions. In: A. Canale, A. Luati, S. Mazzuco, R. Piccarreta, N. Sartori, P. Secchi (Eds.). *Advances in Neural Data Science* (pp. 35–55). Springer. ISBN: 978-3-031-70638-7. DOI: [10.1007/978-3-031-70638-7_3](https://doi.org/10.1007/978-3-031-70638-7_3).

MANUSCRIPT UNDER REVIEW

[1] **Barile, F.** & Nemeth, C. (2024). Control variate-based stochastic sampling from the probability simplex. *Under revision (2026)*. *ArXiv*: [\[link\]](#). Code: [\[link\]](#)

[2] **Barile, F.**, Capozio, S. & Nipoti, B. (2026). Efficient Bayesian inference for multiple network data. *Under revision*. *ArXiv*: [\[link\]](#). Code: [\[link\]](#)

MANUSCRIPT IN PREPARATION

- [1] **Barile, F.**, Borrotti, M., Nipoti, B., Poli, G. & Stingo, F. C. (2026+). Data Integration and Missing Imputation for Joint Bayesian Learning under Multiple Nonignorable Missing Mechanism. *In preparation*.
- [2] **Barile, F.**, & Nipoti, B. (2026+). Dependent Bernstein–Dirichlet priors for multiscale density estimation. *In preparation*.
- [3] **Barile, F.**, Nieto-Barajas, L.E., & Nipoti, B. (2026+). A Network of Dependent Dirichlet Processes. *In preparation*.
- [4] **Barile, F.**, Lunagómez, S., & Nemeth, C. (2026+). Covariate-dependent modeling of hypergraph data. *In preparation*.

PH.D. THESIS

- [1] **Barile, F.** (2025). Advances in Bayesian Methods: Nonparametric Network Modeling and Scalable Computations. Ph.D. thesis, XXXVI cycle [\[link\]](#)

PRESENTATIONS AND SEMINARS

INVITED PRESENTATIONS

- CFE-CMStatistics 2026, HTW Berlin, Dec 12-14, 2026
University of Applied Sciences, Berlin, Germany
- StaTalk 2025, University of Milano-Bicocca, Milan, Italy Jun 13-14, 2025
- ECOSTA 2023, Waseda University, Tokyo, Japan Aug 1-3, 2023

CONTRIBUTED PRESENTATIONS

- BAYSM 2023, Online mode Nov 13-17, 2023

POSTERS

- ISBA 2026, Nagoya, Japan Jun 28 - Jul 3, 2026
- Bayes Comp 2025, Singapore Jun 16-20, 2025
- SISBayes, Padova, Italy Sep 4-5, 2025
- Complex Graphical Models for Biological Network Science, Padova, Italy Sep 4, 2024
- ISBA 2024, Venice, Italy Jul 1-7, 2024
- IMS ICSDS 2022, Florence, Italy Dec 13-16, 2022

SEMINARS

- Seminar in the Computational Statistics and Machine Learning series Feb 2024
School of Mathematical Sciences, Lancaster University, Lancaster, UK

WORKSHOPS

- Data Research Camp 2022, VIU, San Servolo Island, Venice, Italy Jul 12-15, 2022

TEACHING

Lecturer

- Inferential Statistics [ENG] 2026/2027
(Master's degree in International Economics and Master's degree in Global Management)
Department of Economics, Management and Statistics, University of Milano-Bicocca

Adjunct Lecturer

- Industrial Economics and Dynamics [ITA] 2021/2022
(Master's degree in Statistical and Economic Sciences)
Department of Economics, Management and Statistics, University of Milano-Bicocca

Teaching Assistant

• Inferential Statistics [ENG] (Master's degree in International Economics)	2025/2026
Department of Economics, Management and Statistics, University of Milano-Bicocca	
• Inferential Statistics [ENG] (Master's degree in International Economics)	2024/2025
Department of Economics, Management and Statistics, University of Milano-Bicocca	
• Advanced Statistics [ENG] (Master's degree in Marketing and Global Markets)	2024/2025
Department of Economics, Management and Statistics, University of Milano-Bicocca	
• Inferential Statistics [ENG] (Master's degree in International Economics)	2023/2024
Department of Economics, Management and Statistics, University of Milano-Bicocca	
• Advanced Statistics [ENG] (Master's degree in Marketing and Global Markets)	2023/2024
Department of Economics, Management and Statistics, University of Milano-Bicocca	
• Inferential Statistics [ENG] (Master's degree in International Economics)	2022/2023
Department of Economics, Management and Statistics, University of Milano-Bicocca	

SERVICE TO PROFESSION

ORGANIZATION OF SCIENTIFIC EVENTS

Organizer and chair of the Invited Session “Recent advances in Bayesian variable selection and shrinkage” of CFE-CMStatistics 2026 [\[link\]](#)

REFeree SERVICE

Computational Statistics and Data Analysis; Communications in Statistics - Simulation and Computation; IEEE Journal of Biomedical and Health Informatics.

POSITIONS IN ACADEMIC SOCIETIES

Member of ISBA, j-ISBA Section, BNP-ISBA Section

Member of the International Society for Bayesian Analysis (ISBA), Junior ISBA Section, Bayesian non-parametric ISBA Section

Member of SIS, y-SIS

Member of the Italian Statistical Society (Società Italiana di Statistica, SIS), Young SIS

THESIS SUPERVISION

Co-supervision of a Master's thesis, Master's Degree in Statistical and Economic Sciences, Department of Economics, Management and Statistics, University of Milano-Bicocca

COMPUTER SKILLS

Programming languages	R, C++, Python (basic), SQL, Java, $\text{\LaTeX}$
Software	SAS, Stata
	RStudio:
	• C++ based packages creation;
	• parallel programming;
	• create dynamic reports and reproducible research using $\text{\LaTeX}$;
	• project version control via GitHub.

LANGUAGES

Italian	Mother tongue
English	Fluent
French	Basic

OTHER EXPERIENCES

- **Junior Data Scientist**

Apr 2019 - Sep 2020

CRIF S.p.A.

- **Degree Program Tutor**

Feb - Jul 2018

Department of Economics, Statistics and Finance, University of Calabria

- **Degree Program Tutor**

Apr - Oct 2017

Department of Economics, Statistics and Finance, University of Calabria

Le dichiarazioni rese nel presente curriculum sono da ritenersi rilasciate ai sensi degli artt. 46 e 47 del D.P.R. 445/2000. Autorizzo la pubblicazione ai sensi del D.Lgs. n. 33/2013 e il trattamento dei dati personali ai sensi del Regolamento (UE) 2016/679 e del D.Lgs. n. 196/2003.

Milan, October 4, 2026

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